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Module 11:
Electrical machines

Dr.-Ing. Ralf Wegener
Grzegorz Pawel Lisicki
Josef Kirschner

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1 Classification of electrical machines

Electrical machines is another term for electromagnetic energy converters and describes, on the one hand, transformers as stationary electrical machines and, on the other hand, electric motors and generators. Electric motors power many everyday devices such as household appliances and electric vehicles, while generators and transformers ensure the generation and distribution of electrical energy.

Dieses Kapitel bietet eine Einführung in die grundlegenden Konzepte und Funktionsweisen elektrischer Maschinen. Zunächst werden die verschiedenen Typen elektrischer Maschinen vorgestellt, einschließlich ihrer Prinzipien der Energieumwandlung und ihrer Hauptkomponenten. Es folgt eine detaillierte Betrachtung der Funktionsweise von Elektromotoren und Generatoren, der Relevanz von Transformatoren für die Energieübertragung sowie der Einfluss neuer Technologien auf die Effizienz und Leistung elektrischer Maschinen.



Learning objectives: Elektrische Maschinen

The students

- are familiar with the various types of electrical machines.
- can solve basic problems in the field of electrical machines.

Electrical machines are energy converters. A distinction is primarily made between transformers and motors or generators. Transformers convert electrical energy into another form of electrical energy by transferring one voltage level to another. Motors and generators, on the other hand, convert electrical energy into mechanical energy and vice versa. Motors convert electrical energy into mechanical energy, while generators convert mechanical energy into electrical energy. Since the conversion process is reversible, motors can also be used as generators and vice versa. Motors and generators are further classified according to the type of current they operate on: direct current machines and alternating current machines. Alternating current machines are further divided into synchronous and asynchronous machines.

The functioning of all electrical machines is based on the electromagnetic principles of induction and Lorentz force discussed in Module 6. While the transformer is based purely on the principle of induction, the motor or generator uses different electromagnetic principles depending on the conversion process. When functioning as a generator, it relies on induction to generate electrical energy through changing magnetic fields. It utilises the principle of Lorentz force when functioning as a motor.

2 The transformer

The transformer is a stationary electrical machine and serves to convert electrical energy through the magnetic circuit back into electrical energy, particularly changing the voltage level. Transformers are available in various designs and power ratings, from a few watts (Figure 2.2) up to the high megawatt range (Figure 2.1).

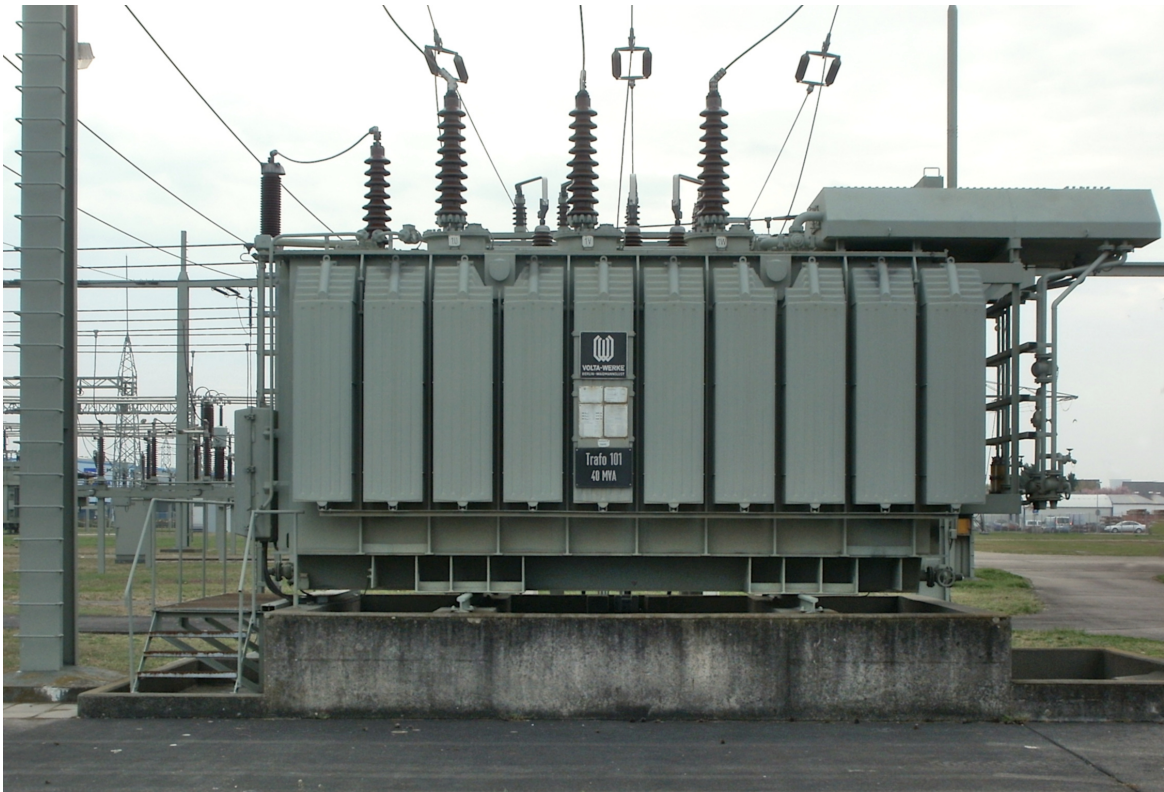


Figure 2.1: **A 40 MVA power transformer.** Recorded at the Altlußheim substation. (Source: Wikimedia, Licence CC0)

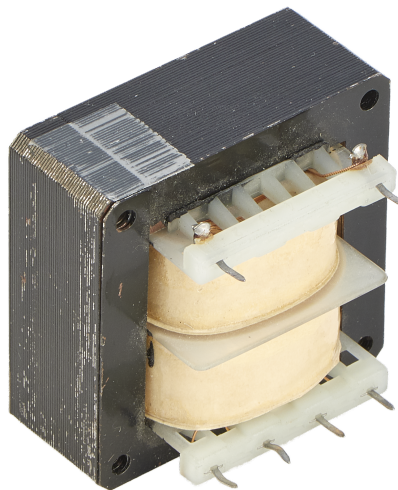


Figure 2.2: **Photograph of a small transformer.**

In public power supply, the energy generated by power plants is transformed to a higher voltage to enable long-distance transmission. Transmission losses decrease with higher voltage, and thus, for the same power, the current decreases quadratically (Equation 2.1 in three-phase alternating current).

$$P_v = 3 \cdot R_L \cdot I_L^2 \quad (2.1)$$

The transmitted power in three-phase alternating current is:

$$\underline{S}_N = \sqrt{3} \cdot \underline{U}_N \cdot \underline{I}_L^* \quad (2.2)$$

The losses are therefore proportional to $\frac{1}{\underline{U}_N^2}$

A transformer makes it possible to provide comprehensive energy supply, as power stations can also be located far away from consumers.

In power supply networks (see Figure 2.3), there are several voltage levels. The extra-high voltage level, which in Europe is predominantly 380 kV, serves as the transmission network for transporting electrical power over long distances, i.e., several hundred kilometers, even across national borders. Large power plants are directly connected at this level. The high-voltage level (110 kV or 220 kV) is referred to as the distribution network and is used to connect cities and large industrial consumers. The medium-voltage level (5–35 kV) provides the connection of local distribution transformers for supplying individual streets. While the higher voltage levels are usually implemented using overhead lines, the medium-voltage network is predominantly realized with underground cables. In rural areas, however, overhead lines are also used at this level.

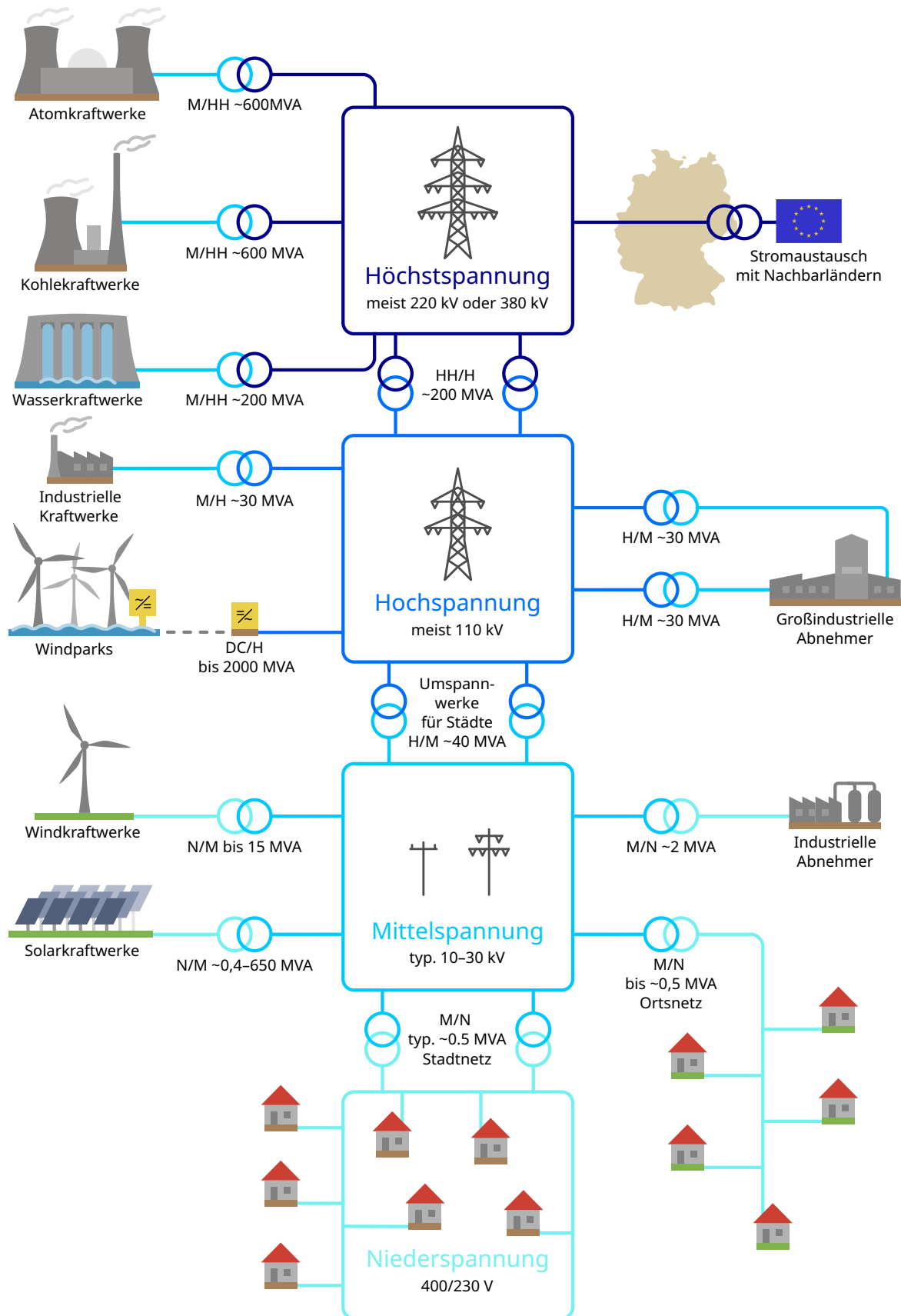


Figure 2.3: Structure of the public electricity supply.

Das Stromnetz ist in mehrere Spannungsebenen unterteilt, die den Transport und die Verteilung der Energie ermöglichen.

2.1 Principle of the transformer

To discuss the principle of the transformer, we start with a completely lossless, ideal transformer. A sinusoidal (actually cosinusoidal, but this is the same shape and only corresponds to a phase shift of 90°) voltage $\underline{U}_1$ is applied to the primary coil. The time curve of $\underline{U}_1$ corresponds to u in Equation 2.3.

$$u = \hat{U} \cdot \cos(\omega t) \quad (2.3)$$

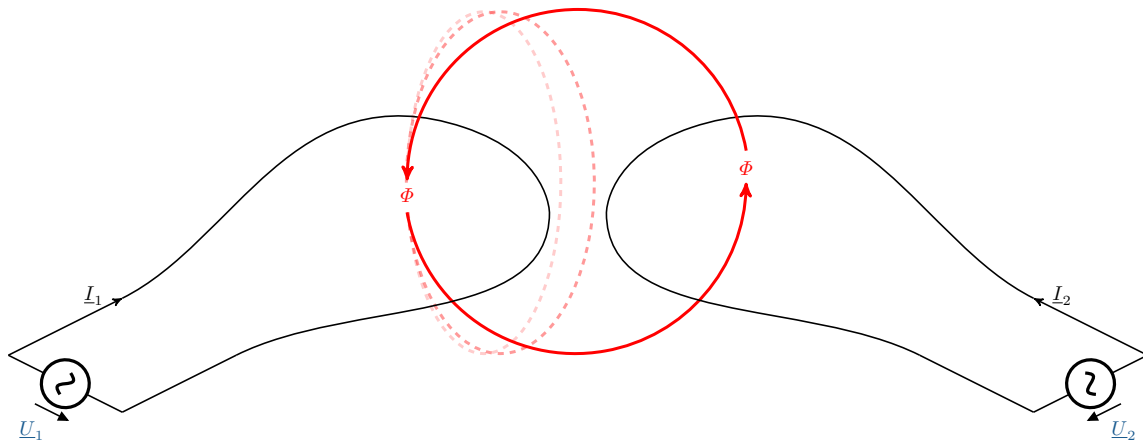


Abbildung 2.4: **Principle of magnetic coupling in a transformer.** The voltage $\underline{U}_1$ in the left conductor loop generates a magnetic flux Φ , which induces the voltage $\underline{U}_2$ in the right conductor loop.

According to the inverse of the law of induction (equation 2.4), this voltage causes a changing magnetic flux.

$$u = -N \cdot \frac{d\Phi}{dt} \quad (2.4)$$

The shape of the magnetic flux is also sinusoidal for a sinusoidal voltage waveform, since the integral of a cosine is a sine. The effective value of the magnetic flux can therefore be represented by equation 2.5 by solving the induction law for Φ and forming the effective value. This eliminates the sine and thus the time dependence.

$$\Phi = \frac{\sqrt{2} \cdot \underline{U}_1}{2\pi \cdot N_1 \cdot f} \quad (2.5)$$

The direction of the magnetic field always moves according to the right-hand rule (see Module 4), in this example counterclockwise. The magnetic flux generated in this way induces a voltage $\underline{U}_2$ in the secondary coil of the transformer, which is optimally penetrated by the same flux Φ . This voltage can also be calculated using the law of induction (equation 2.4).

$$u_2 = -N_2 \cdot \frac{d\Phi}{dt} \quad (2.4)$$

The applied primary voltage $\underline{U}_1$ and the secondary voltage $\underline{U}_2$ can be put into a ratio (Equation 2.6). The rate of change of the magnetic flux Φ is thereby eliminated, leaving only the number of

turns on the primary side N_1 and the secondary side N_2 . This ratio is called the transformation ratio $\ddot{u}$ (equation 2.6). If the transformation ratio $\ddot{u}$ is greater than one, the voltage is reduced (step-down transformer). If the transformation ratio $\ddot{u}$ is less than one, the voltage is increased (step-up transformer). Since the transformer can be used in both directions, a transformation ratio greater than or equal to one is always specified in the technical data.

$$\frac{U_1}{U_2} = \frac{-N_1 \cdot \frac{d\Phi}{dt}}{-N_2 \cdot \frac{d\Phi}{dt}} = \frac{N_1}{N_2} = \ddot{u} \quad (2.6)$$

$\ddot{u} < 1 \rightarrow$ Voltage goes up

$\ddot{u} > 1 \rightarrow$ Voltage goes down

The following is always specified $\ddot{u} > 1$

In a real transformer, not all of the magnetic flux generated by the primary winding penetrates the secondary side (shown as a red dotted line in Figure 2.4). These losses are called stray flux and are represented in the equivalent circuit diagram by the stray inductance. Therefore, the real secondary voltage is smaller than the ideal secondary voltage calculated here.

2.2 Transformer structure

In simple terms, a transformer consists of two coils that are coupled together by means of a magnetically conductive core (see Figure 2.5). The core is made of a ferromagnetic material. Due to its high permeability, the ferromagnetic material has the ability to conduct the magnetic flux between two coils as efficiently as possible (see Module 6). However, effects such as hysteresis and eddy current losses interfere with the resistance-free flow. These effects are referred to as iron losses.

The coils are wound around the magnetic conductor. As mentioned above, the number of turns determines the voltage transformation ratio (equation 2.6). They consist of electrically conductive wires such as copper. In reality, these also exhibit resistance losses, which are also referred to as copper losses.

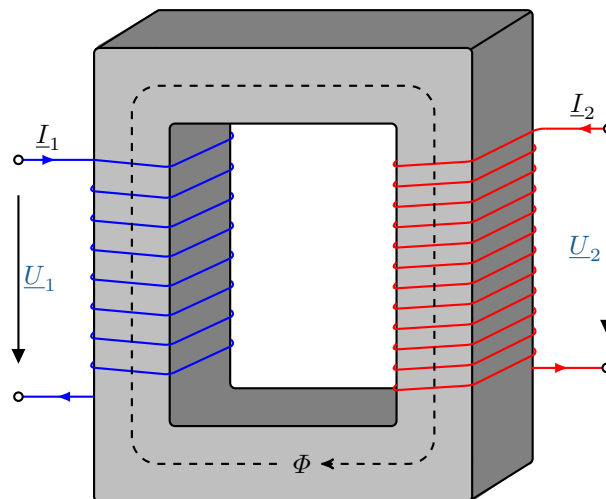


Figure 2.5: **Schematischer Aufbau des Transformators.** Zwei Spulen sind mittels eines ferromagnetischen Kerns miteinander gekoppelt. Die Windungszahlen bestimmen dabei das Übersetzungsverhältnis der Spannung.

In the circuit diagram of the ideal transformer, the magnetically coupled coils are represented as adjacent black rectangles (see Figure 2.6). The two lines between the coils indicate that they are connected by a continuous magnetically conductive core. The dots on the respective coils provide information about the winding direction. They can be located at the bottom or at the top. If they are at the same height, the phase position of the two coils is the same. If they are at different heights, this means that the winding direction is reversed, resulting in a phase shift of 180° .

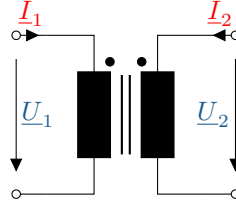


Figure 2.6: Ideal circuit diagram of a transformer.

Example 2.1: Ideal three-phase mains transformer

An ideal three-phase power transformer has an apparent power of $\underline{S}_N = 100 \text{ kVA}$, an upper-side voltage of $\underline{U}_1 = 20 \text{ kV}$ (delta connection) and a lower-side voltage of $\underline{U}_2 = 400 \text{ V}$ (delta connection).

- a) What is the gear ratio?

$$\begin{aligned} \dot{i} &= \frac{\underline{U}_1}{\underline{U}_2} = \frac{N_1}{N_2} \\ &= \frac{20 \text{ kV}}{400 \text{ V}} = 50 \end{aligned}$$

- b) What are the primary and secondary currents in nominal operation at a power factor of $\cos(\varphi) = 1$?

$$\begin{aligned} \cos(\varphi) = 1 &\rightarrow |\underline{S}| = P \\ P &= U \cdot I \\ I_1 &= \frac{P}{U} = \frac{100 \text{ kW}}{20 \text{ kV}} = 5 \text{ A} \\ I_2 &= \frac{P}{U} = \frac{100 \text{ kW}}{400 \text{ V}} = 250 \text{ A} \end{aligned}$$

2.3 Euivalent circuit diagram

For better understanding, the equivalent circuit of the real transformer (Figure 2.7) is first represented functionally as a galvanically isolated system with a primary and a secondary circuit. Compared to the ideal transformer, this equivalent circuit is extended by the losses described previously. The magnetic circuit is transformed into an electrical circuit using the main inductance L_h and the leakage inductances $L_{\sigma 1}$ and $L_{\sigma 2}$. The copper losses in the primary and secondary windings described earlier are represented by the resistances R_1 and R_2 . The quantity $\underline{I}_\mu$ denotes the magnetizing current and describes the component of the no-load current required to excite the magnetic core of the transformer. For simplicity, iron losses are neglected in this example. If they are to be taken into account, the iron losses are represented by a resistor connected in parallel with the main inductance.

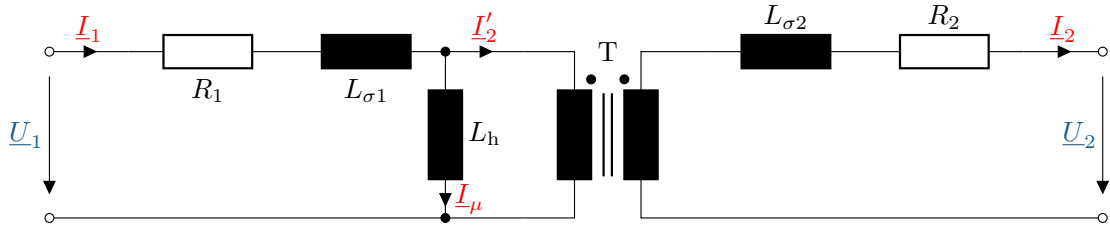


Abbildung 2.7: Equivalent circuit diagram of a galvanically isolated transformer.

Subsequently, in the T-equivalent circuit of the transformer (Figure 2.8), it is assumed that there is no galvanic isolation between the primary and secondary sides, although in reality this is of course the case. This step allows the quantities on the secondary side to be referred to the primary side and the calculation rules known from Module 4 to be applied. The referred quantities are denoted as primed quantities, in this example as $\underline{U}'_2$, $\underline{I}'_2$, R'_2 , and $L'_{\sigma 2}$.

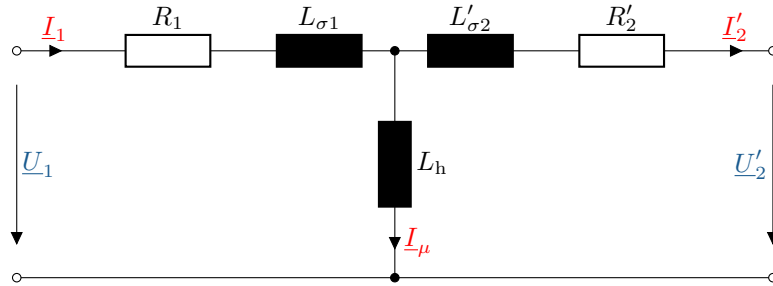


Abbildung 2.8: T-replacement circuit diagram of a transformer.

To convert the voltage to the primary side, the voltage $\underline{U}'_2$ is multiplied by the transformation ratio. To calculate the incoming current $\underline{I}_1$, the current $\underline{I}_2$ is multiplied by the inverse transformation ratio and then added to the magnetising current $\underline{I}_\mu$ (Equation 2.8).

$$\underline{U}_1 = \underline{U}'_2 = \frac{N_1}{N_2} \cdot \underline{U}_2 \quad (2.7)$$

$$\underline{I}_1 = \underline{I}'_2 + \underline{I}_\mu = \frac{N_2}{N_1} \cdot \underline{I}_2 + \underline{I}_\mu \quad (2.8)$$

The losses occurring on the secondary side $L'_{\sigma 2}$ and R'_2 are converted to the primary side using the squared transmission ratio (Equations 2.9 and 2.10).

$$L'_{\sigma 2} = \left(\frac{N_1}{N_2} \right)^2 \cdot L_{\sigma 2} \quad (2.9)$$

$$R'_2 = \left(\frac{N_1}{N_2} \right)^2 \cdot R_2 \quad (2.10)$$



Example 2.2: Real three-phase mains transformer

A real three-phase mains transformer ($f = 50 \text{ Hz}$) has the following data:

- Power: $\underline{S}_N = 100 \text{ kVA}$

- Overvoltage $\underline{U}_1 = 20 \text{ kV}$
- Transmission ratio: 50
- $L_h = 500 \text{ H} \Rightarrow \underline{Z}_{Lh} = j157,08 \text{ k}\Omega$
- $L_{\sigma 1} = L'_{\sigma 2} = 190 \text{ mH} \Rightarrow \underline{Z}_{L\sigma} = j59,69 \Omega$
- $R_1 = R'_2 = 30 \Omega$

a) What is the open-circuit voltage $\underline{U}_2$ for this transmission ratio?

$$\begin{aligned}\underline{U}'_2 &= \underline{U}_1 \cdot \frac{\underline{Z}_h}{\underline{Z}_h + \underline{Z}_{\sigma 1} + R_1} \\ &= 19,992 \text{ kV} \cdot e^{j0,011^\circ} \\ \underline{U}_2 &= 399,85 \text{ V} \cdot e^{j0,011^\circ}\end{aligned}$$

b) How high is the no-load current (on the primary side)?

$$\begin{aligned}\underline{I}_1 &= \frac{\underline{U}_1}{\underline{Z}_h + \underline{Z}_{\sigma 1} + R_1} \\ &= \frac{20 \text{ kV}}{157,139 \text{ k}\Omega \cdot e^{j89,989^\circ}} \\ &= 127,28 \text{ mA} \cdot e^{-j89,989^\circ}\end{aligned}$$

c) How significant are the idle losses?

$$\begin{aligned}\underline{S}_v &= \underline{U} \cdot \underline{I}^* \\ &= 20 \text{ kV} \cdot (127,28 \text{ mA} \cdot e^{-j89,989^\circ})^* \\ &= 2,546 \text{ kVA} \cdot e^{j89,989^\circ}\end{aligned}$$

3 Direct current machine

DC machines are electric machines that operate on direct current and can be used both as motors and generators. They are characterised by precise controllability of speed and torque, making them ideal for applications with variable speed and load. DC machines have high starting torque and allow easy reversal of the direction of rotation. They are often used in electric vehicles, industrial control systems and battery-powered devices.

Advantages:

- Simpler and more cost-effective
Design of the power converter
- High control dynamics
- Direct operation of the machine with accumulators possible
- High overload capacity

Disadvantages:

- High construction costs
- High maintenance (brushes)
- Low power density
- Expenses

3.1 Digression: Principle of the electric motor

First, the principles presented in Module 6 will be reviewed. Figure 3.1 shows three diagrams that summarise how an electric motor works. The basic structure of an electric motor consists of a stationary part, called the stator, and a rotating part, called the rotor. The three figures show a conductor coil in a horseshoe magnet. The horseshoe magnet forms the stator and a conductor coil that can move around its own axis represents the rotor. The first diagram illustrates the isolated course of the magnetic field lines of the horseshoe magnet. They run vertically from the north to the south pole within the horseshoe magnet. The second diagram shows the isolated course of the magnetic field lines of the current-carrying conductor loop. The flow of current creates two opposing eddy fields at the conductor loop. The third diagram shows the interaction between the horseshoe magnet (stator field) and the current-carrying conductor loop (rotor field). The two fields overlap. The Lorentz force resulting from the overlap causes the conductor loop to rotate according to the right-hand rule. Electromagnetic energy is converted into mechanical energy.

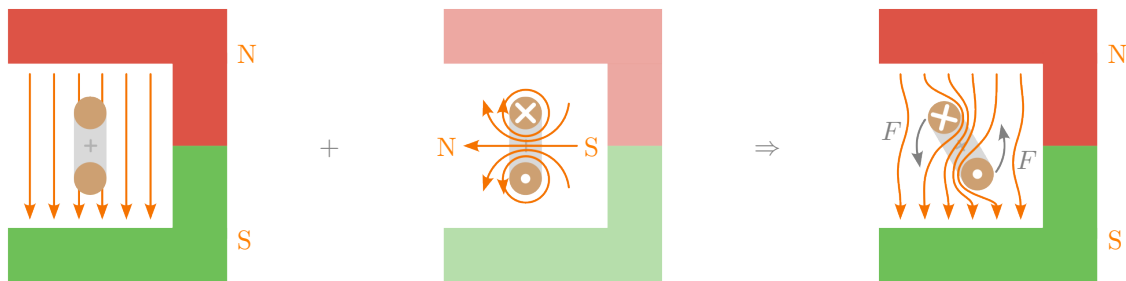


Figure 3.1: **Principle of the electric motor.** The Lorentz force causes the current-carrying conductor loop in the magnetic field to rotate.

The Lorentz force can be calculated using the formula familiar from Module 6:

$$F = B \cdot I \cdot \ell \cdot N \quad (3.1)$$

Example 3.1: Power of a direct current machine

A direct current motor has a magnetic flux density of $B = 0.8 \text{ T}$ in the air gap. There are a total of $N = 400$ armature wires under the poles, through which a current of $I = 10 \text{ A}$ flows. The effective conductor length is $\ell = 150 \text{ mm}$.

Calculate the force F at the circumference of the anchor.

$$\begin{aligned} F &= B \cdot I \cdot \ell \cdot N \\ &= 0,8 \frac{\text{Vs}}{\text{m}^2} \cdot 10 \text{ A} \cdot 0,15 \text{ m} \cdot 400 \\ &= 480 \frac{\text{kg} \cdot \text{m}^2 \cdot \text{s} \cdot \text{A} \cdot \text{m}}{\text{s}^3 \cdot \text{A} \cdot \text{m}^2} = 480 \text{ N} \end{aligned}$$

3.2 Structure and housing design

The construction of the DC machine is illustrated using three figures. Figure 3.2 shows a diagonal sectional model of the DC machine, providing a general overview of the components of the stator and the rotor. For a more detailed illustration of the structure, Figure 3.3, focusing on the stator,

and Figure 3.4, illustrating the rotor of a DC machine, are used.

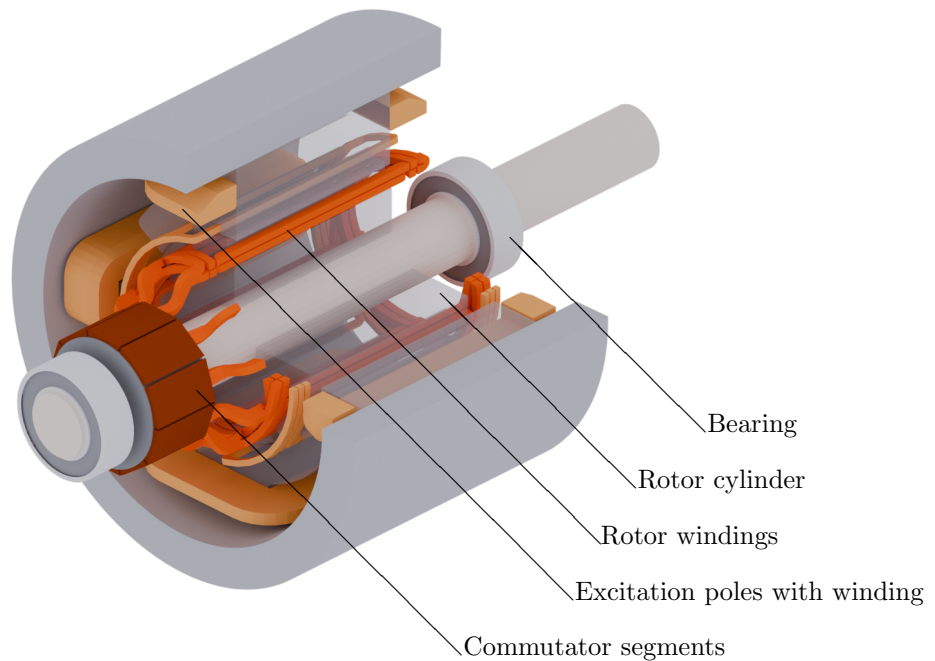


Figure 3.2: **Diagram of a cross-section of a direct current machine.**

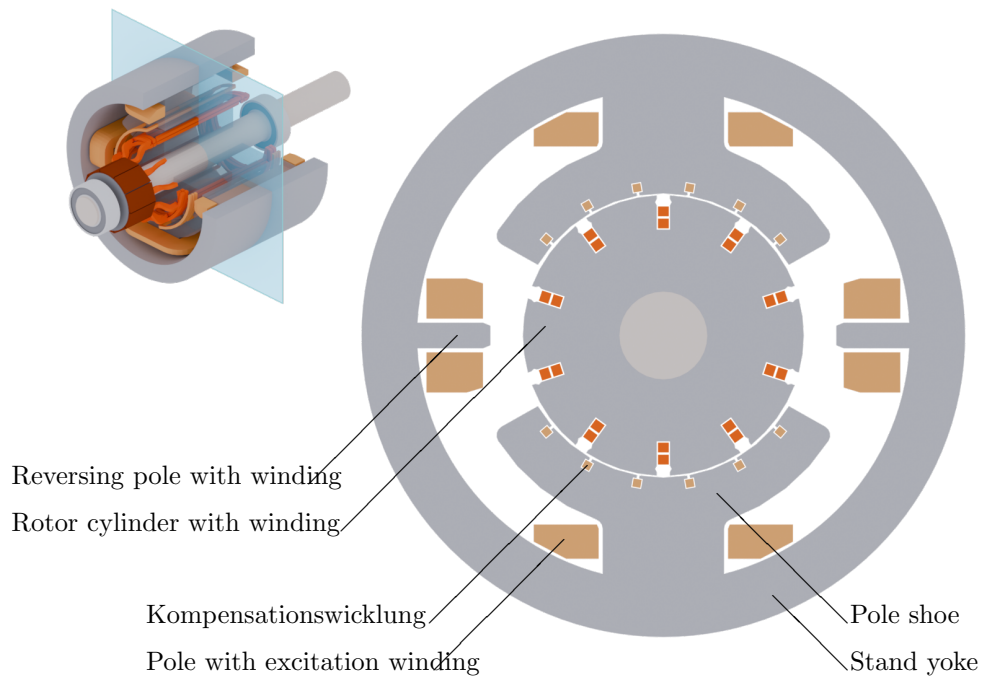


Figure 3.3: **Basic structure of a direct current machine. Here with only one pole pair.** (Pole pair number $p = 1$).

The components of the stator, also called the stationary part, are clearly illustrated in the frontal cross-section of a DC machine (Figure 3.3). One of the fundamental components is the field windings

, which, wound around a pole shoe, form a magnetic pole. The poles are responsible for generating a magnetic field, known as the excitation field.

Since a magnetic field always consists of a north and a south pole, the poles are installed in opposing pairs. The resulting characteristic quantity, the number of pole pairs, provides information about the operating characteristics of the DC machine. A higher number of pole pairs reduces the rotational speed but increases the torque. In very small machines, the field windings can be replaced by permanent magnets. In larger machines (approximately above 1 kW), additional interpole windings are present; however, these are neglected here.

In Figure 3.4, the components of the rotor are clearly visible. In general, the rotor is also referred to as the armature or the rotating part. The function of the rotor is to convert electromagnetic energy into mechanical energy by rotation about its own axis (and vice versa). For this purpose, the rotor is mounted on a motor shaft, which transmits the mechanical energy. The ball bearing at the end of the motor shaft reduces friction losses during energy transmission. To set the rotor in motion, the interaction between the commutator and the rotor winding is essential. The rotor winding, also called the armature winding, is responsible for generating a magnetic field, known as the armature field. Only through the interaction of the armature field with the excitation field is a torque exerted on the rotor, causing it to rotate. The laminated cores around which the coils are wound consist of insulated steel sheets in order to reduce eddy currents in the windings. The commutator, also called current reverser, has the function of supplying the rotor winding with current in a commutated manner. It consists of several mutually insulated segments (lamellae), each of which is connected to one branch of the rotor winding. The current is supplied to the commutator segments via carbon brushes mounted on the stator. Due to the rotation of the segments beneath the brushes, the commutator acts as a mechanical switch and ensures that the direction of current in the rotor windings located under the main poles remains constant.

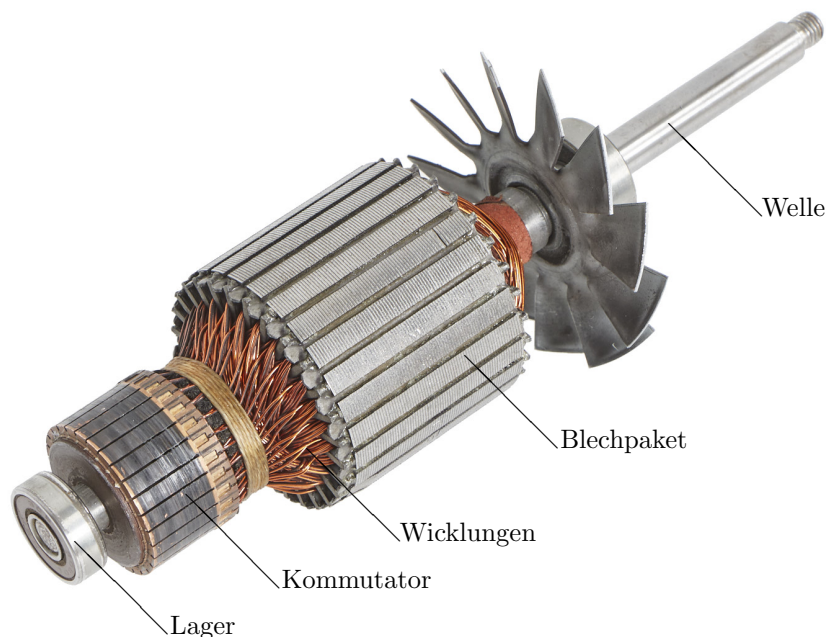


Figure 3.4: **Photo of a rotor.** The characteristic structural elements can be seen here.

3.3 Magnetic fields

The magnetic field in which the conductor moves is decisive for the Lorentz force. This field is generated by the stator with the excitation windings and is approximately parallel and homogeneous

in the area of the rotor.

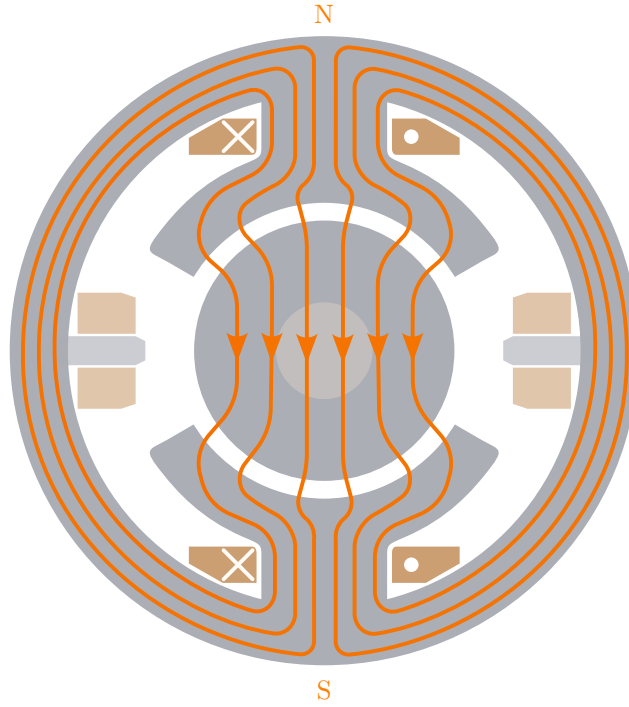


Figure 3.5: **Excitation field of a direct current machine.** Schematic representation in front view.

3.4 Calculation of torque

In Module 6 and Chapter 3.1, the force acting on a current-carrying conductor in a magnetic field has already been explained. To calculate the torque of the DC machine, the induced voltage is first determined (see Module 6). When the rotor of an excited DC machine rotates, a voltage is induced in each conductor loop of the armature. According to the law of induction (Equation 2.4), the magnitude of the induced voltage (source voltage) U_q depends on the magnetic flux Φ and its time variation, in this case the rotational speed n or the associated angular frequency $\omega = 2\pi \cdot n$. If the specific geometric properties of the rotor are additionally taken into account in the form of the armature constant K , then the product of these characteristic quantities yields the induced voltage U_q of the DC machine (see Equation 3.2).

$$U_q = K \cdot \Phi \cdot \omega \quad (3.2)$$

If both electrical and mechanical losses are neglected, the induced voltage can also be specified directly via the so-called internal power P_i . This then corresponds to both the electrical power $U_q \cdot I_A$ and the mechanical power P_{mech} , which is calculated from the product of the internal torque M_i and the angular velocity ω .

$$P_i = U_q \cdot I_A = P_{\text{mech}} = M_i \cdot \omega \quad (3.3)$$

If this finding is now inserted into equation 3.2, the internal torque M_i is calculated from the product of the armature constant K , the magnetic flux Φ and the armature current I_A

$$M_i = K \cdot \Phi \cdot I_A \quad (3.4)$$

In motor operation, the torque at the machine shaft is the internal torque M_i minus the losses M_V . The machine torque can also be expressed by the efficiency η_1 . However, this efficiency disregards

the electrical losses due to the armature resistance and excitation.

$$M = M_i - M_V \quad M = \eta_1 \cdot M_i \quad (3.5)$$

- Loss moment: M_V [Nm]
- Efficiency: η [1]

3.5 Externally excited DC machine

The excitation circuit in the stator of the DC machine and the rotor circuit, which is supplied with current via the commutator, are in principle independent of each other. If both circuits are supplied by different voltage sources, the machine is referred to as separately excited. The equivalent circuit is shown in Figure 3.6.

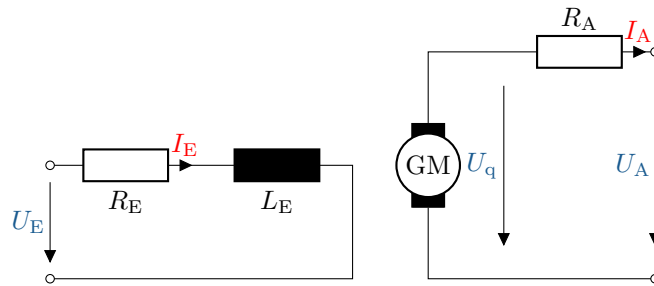


Figure 3.6: Equivalent circuit diagram of a separately excited DC machine.

The excitation circuit consists of the inductance of the excitation winding L_E and the copper resistance R_E of the coil. In the armature circuit, the winding is represented as a circuit with indicated slip rings, as it represents both an inductance and a voltage source. The armature also has an ohmic resistance R_A .

The mesh circulation in the rotor circle results in:

$$U_A = U_q + I_A \cdot R_A \quad (3.6)$$

Substituting equations 3.2 and 3.4 yields the speed/torque and armature current characteristic curve for the separately excited DC machine:

$$n = \frac{U_A}{2\pi K \cdot \Phi} - \frac{R_A \cdot M_i}{2\pi(K \cdot \Phi)^2} \quad (3.7)$$



Example 3.2: Externally excited direct current machine

A separately excited DC machine with the armature constant $K = \frac{1}{2\pi}$ has a no-load speed of $n = 1200 \frac{1}{\text{min}}$ at an armature voltage of $U_A = 400 \text{ V}$. The armature resistance is $R_A = 2.3 \Omega$.

- a) How large is the pathogen flow? Φ ?

At idle speed, the internal torque M_i is equal to zero. Equation 3.7 is therefore simplified

to:

$$n = \frac{U_A}{2\pi K \cdot \Phi}$$

$$\Phi = \frac{U_A}{2\pi K \cdot n} = \frac{400 \text{ V}}{\frac{2\pi}{2\pi} \cdot 1200 \frac{1}{\text{min}} \cdot \frac{1}{60} \frac{\text{s}}{\text{min}}} = 20 \text{ Vs}$$

b) How fast does the machine rotate at an internal torque of $M_i = 10 \text{ Nm}$?

$$n = \frac{U_A}{2\pi K \cdot \Phi} - \frac{R_A \cdot M_i}{2\pi (K \cdot \Phi)^2}$$

$$= \frac{400 \text{ V}}{\frac{2\pi}{2\pi} \cdot 20 \text{ Vs}} - \frac{2,3 \Omega \cdot 10 \text{ Nm}}{2\pi (\frac{1}{2\pi} \cdot 20 \text{ Vs})^2}$$

$$= 20 \frac{1}{\text{s}} - 0,362 \frac{1}{\text{s}} = 19,638 \frac{1}{\text{s}} = 1178,3 \frac{1}{\text{min}}$$

c) How big is the anchor current?

$$I_A = \frac{M_i}{K \cdot \Phi} = \frac{10 \text{ Nm}}{\frac{1}{2\pi} \cdot 20 \text{ Vs}} = 3,61 \text{ A}$$

3.6 Series closure machine

In a series-wound motor, the field circuit and rotor circuit are connected in series. The field current is therefore equal to the armature current. This type of motor is often used in simple electrical appliances with alternating current and is therefore also called a universal motor. As a direct current motor, it was previously used primarily in traction drives, for example in trams. Nowadays, however, these have been replaced by three-phase drives with converters due to their higher efficiency.

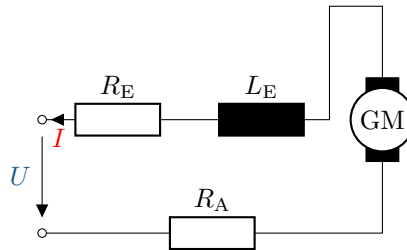


Figure 3.7: Equivalent circuit diagram of a series-wound machine.

- Can also be used for alternating voltage
- high tightening torque

Unlike externally excited DC machines, whose speed-torque characteristic curve is a straight line according to equation 3.7, series-wound machines behave non-linearly. Without load, the speed is theoretically infinite, meaning that the machine will accelerate until it destroys itself through centrifugal force. The machine „goes through“. However, in traction drive applications, idling never occurs; instead, a high starting torque is desired.

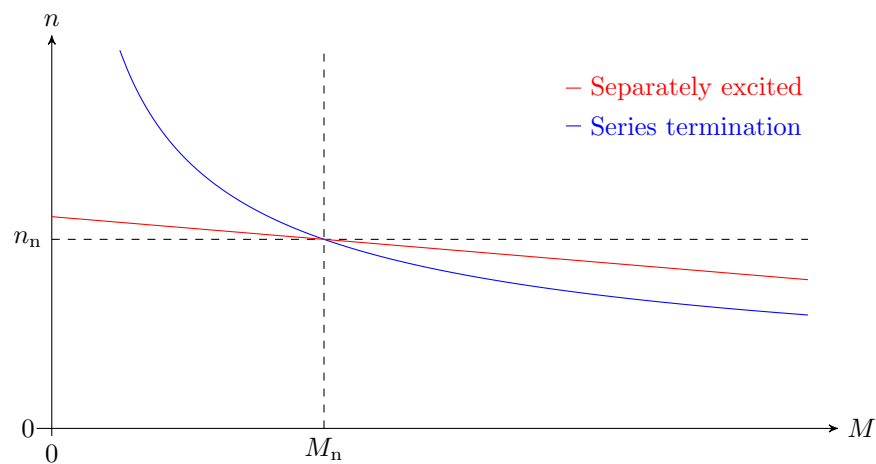


Figure 3.8: **Speed-torque characteristics.** Characteristic curve of a separately excited DC machine and a series-wound machine.

4 synchronous machine

The synchronous machine is a rotating field machine, which is therefore operated with a three-phase alternating current (see Module 7) in the stator circuit. The rotor can be excited either permanently (with permanent magnets) or by a direct current circuit. This type of machine is used in power stations to generate electricity, but as it is highly efficient, it is also increasingly used in modern drive technology in smaller power classes. However, a major disadvantage is that the synchronous machine must always be controlled by power electronics, as it cannot be started up on a simple three-phase network.

Advantages:

- Robust and maintenance-free.
- High control dynamics.
- High power density with low construction volume.
- Overload capacity.

Disadvantages:

- Complex converter technology.
- Complex control system.
- Controlled operation only possible with great effort or not possible at all.

4.1 Structure

- Three stator windings offset by 120° from each other.
- Laminated stator design, as it is subjected to an alternating magnetic field.
- The exciter winding is located in the rotor.

- The rotor is partially unlaminated.

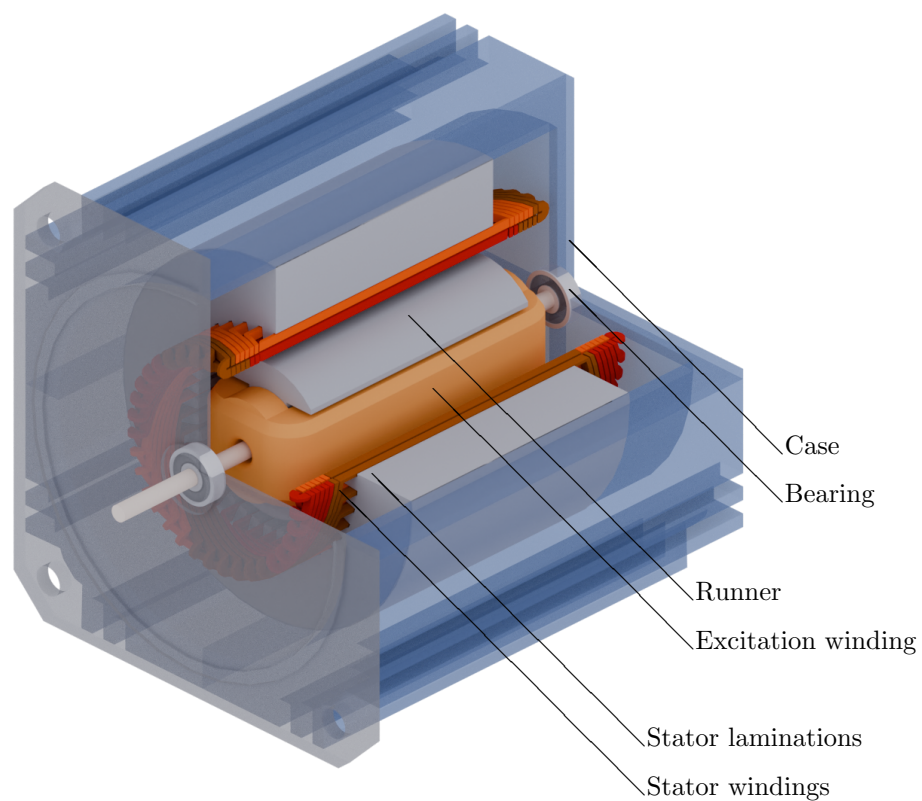


Figure 4.1: **Diagram of a cross-section of an externally excited synchronous machine.**

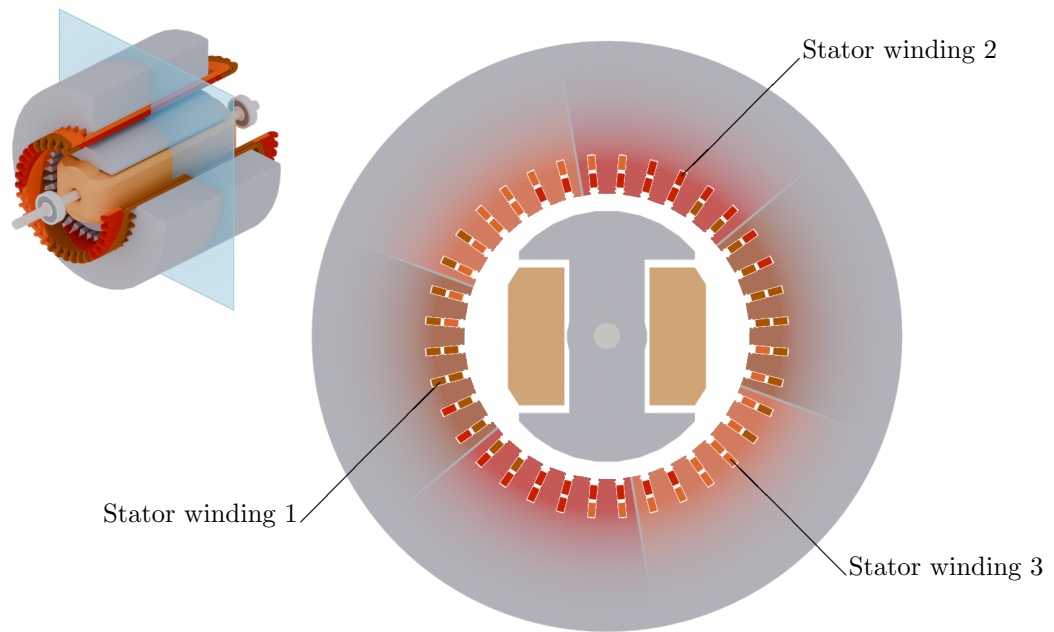


Figure 4.2: **Structure of a separately excited synchronous machine.** The stator windings, offset by 120° from each other, are illustrated here in colour.

4.2 Field gradients

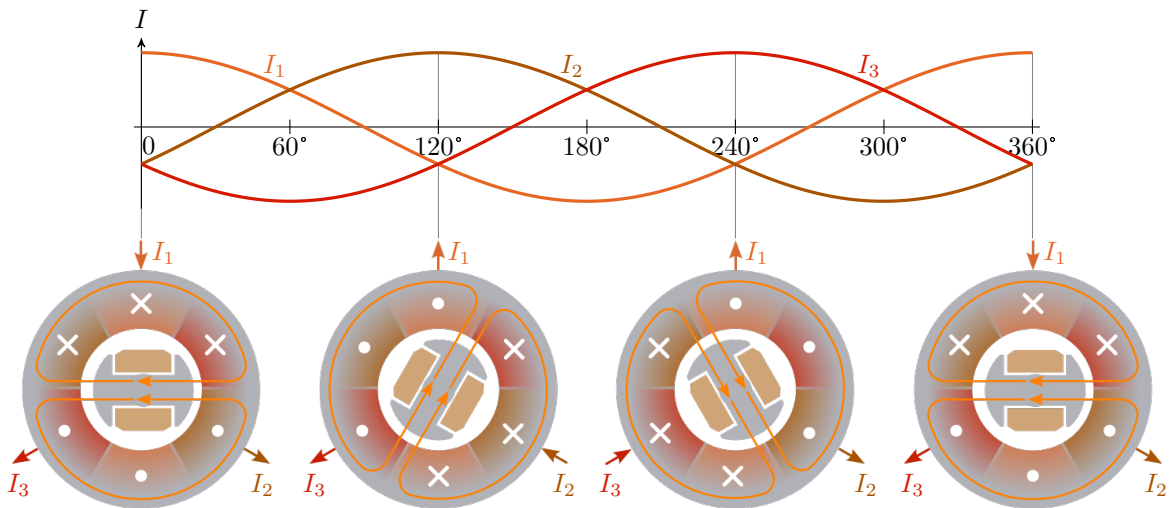


Figure 4.3: **Direction of the rotating field vector.** The rotor rotates synchronously with the sinusoidal rotating fields of the stator, which are offset by 120 degrees.

In a synchronous machine, the magnetic coupling between the stator and rotor generates the torque. In motor operation, the rotor field follows the stator field. The mechanical speed must therefore always be equal to the electrical rotational frequency. Between the stator and the rotor is the load-dependent pole angle ϑ . The torque is greatest at $\vartheta = 90^\circ$. If, during motor operation, the pole wheel angle exceeds 90° due to excessive load, the torque generated decreases and the machine „stalls“—that is, it comes to an abrupt halt.

- The stator flux is created by the superposition of the fluxes of the three windings.
- The current flow is generated by the excitation winding.

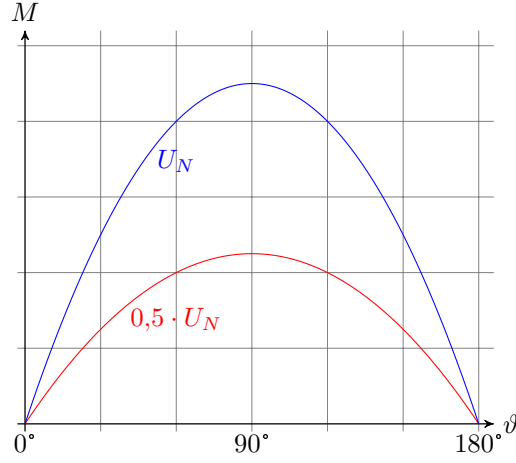


Figure 4.4: **Torque characteristic curve as a function of pole wheel angle.**

4.3 Equivalent circuit diagram

In mains operation, the terminal voltage U_1 and the frequency ω are fixed. This is a common operating condition for synchronous machines.

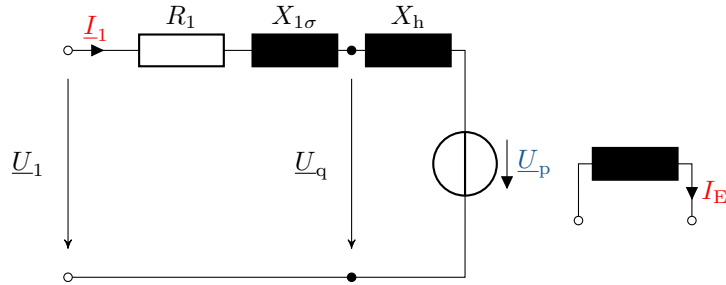


Figure 4.5: **Complex single-phase ESB of a full-pole generator.** For example, in grid operation.

5 Asynchronous machine

The asynchronous machine is also a rotating field machine. The stator is identical to the stator of the synchronous machine, whereas the rotor is passive in the most common version of this machine type, the so-called squirrel cage rotor.

5.1 Aufbau

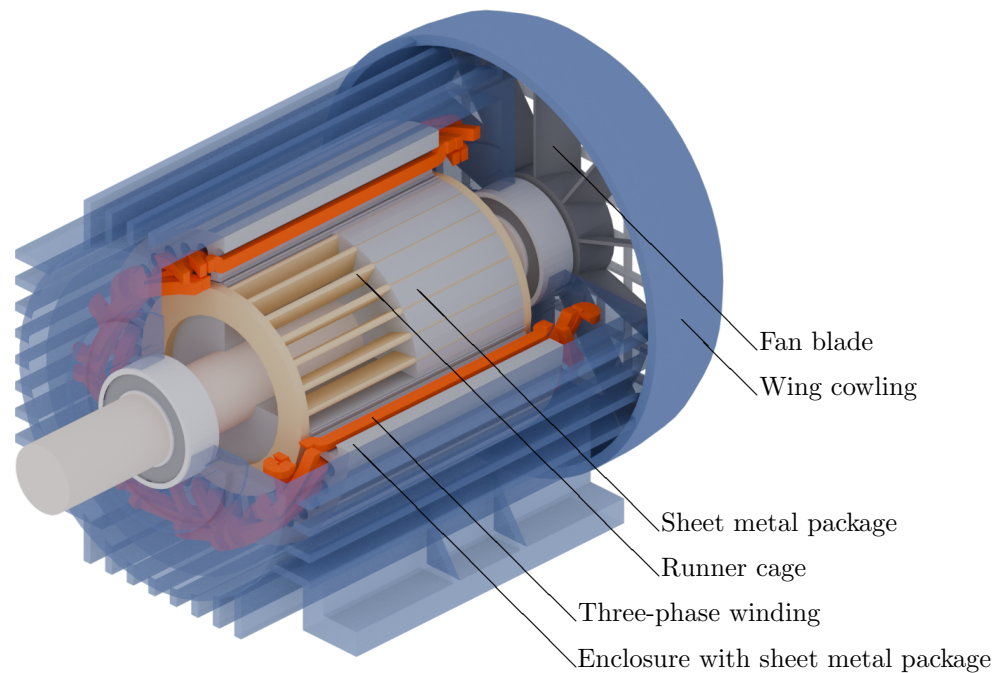


Figure 5.1: **Structure of a cage rotor asynchronous machine.** For clarity, half of the rotor cylinder plates have been cut away to reveal the cage.

5.2 Cage rotor

- Conductive rods are located in the grooves of the runner plate package.
- The rods are connected to each other at the front by short-circuit rings.
- Together, these components form a cage (similar to a hamster cage).
- The rods are designed at an angle to reduce harmonics in the rotating magnetic field.

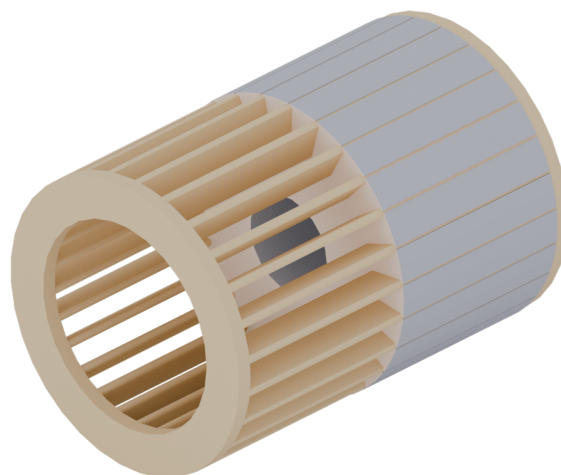


Abbildung 5.2: **Cage of an asynchronous rotor.** Shown with half of the sheet metal package.

5.3 Start-up of the ASM

- The stator generates a rotating magnetic traveling wave in the stationary part with the angular velocity: $\omega_s = \frac{\omega_0}{p}$.
- The stationary rotor experiences a time-varying field with the frequency ω_s .
- The rotating magnetic field penetrates the rotor and induces a voltage with the frequency ω_s .
- The induced voltage results in a current flow, since the rotor windings are short-circuited.
- The induced rotor current opposes the change of the stator field perceived by it (Lenz's law).
- The stator field and the rotor current interact via the Lorentz force, leading to the formation of a torque.
- The torque acts in the direction of the stator's rotating magnetic field.
- The rotor begins to rotate.

5.4 Operation of the ASM

- As the rotor speed increases, the rotor perceives an increasingly slower change of the stator field: $\omega_r = \omega_s - \omega_{\text{mech}}$.
- With increasing speed, both the magnitude and the frequency of the induced voltage in the rotor decrease.
- The magnitude of the torque decreases.
- When the rotor and the stator field rotate at the same frequency, synchronous speed is reached.
- The rotor windings no longer experience a change in the magnetic field.
- The induced current and the torque of the machine become zero.
- Due to friction effects, the rotor is slowed down again, leading to the renewed formation of torque.
- In the steady-state condition, a speed slightly below the synchronous speed is established.

5.5 Speed/torque characteristic curve

The speed/torque characteristic curve of an asynchronous machine is non-linear and can be divided into several sections. The characteristic curve is often plotted as shown in Figure 5.3 using the slip s . The slip refers to the percentage deviation of the mechanical rotational speed of the rotor n_r from the electrical rotational speed n_s of the supplying network in the stator.

$$s = \frac{\omega_s - \omega_r}{\omega_s} \quad (5.1)$$

A slip of $s = 1$ (100%) therefore corresponds to the machine being at a standstill or a speed of $n = 0$, while a slip of $s = 0$ corresponds to mains-synchronous operation, which can never occur with an asynchronous machine, however, as the torque would be zero in this case. The speed/torque characteristic curve is also known as the Klosssche characteristic curve and can be calculated using the equation 5.2 ausgedrückt werden:

$$M = \frac{2 \cdot M_K}{\frac{s_K}{s} + \frac{s}{s_K}} \quad (5.2)$$

For large machines, the slip corresponds very closely to the percentage losses of the machine:

$$\eta \approx 1 - s \quad (5.3)$$

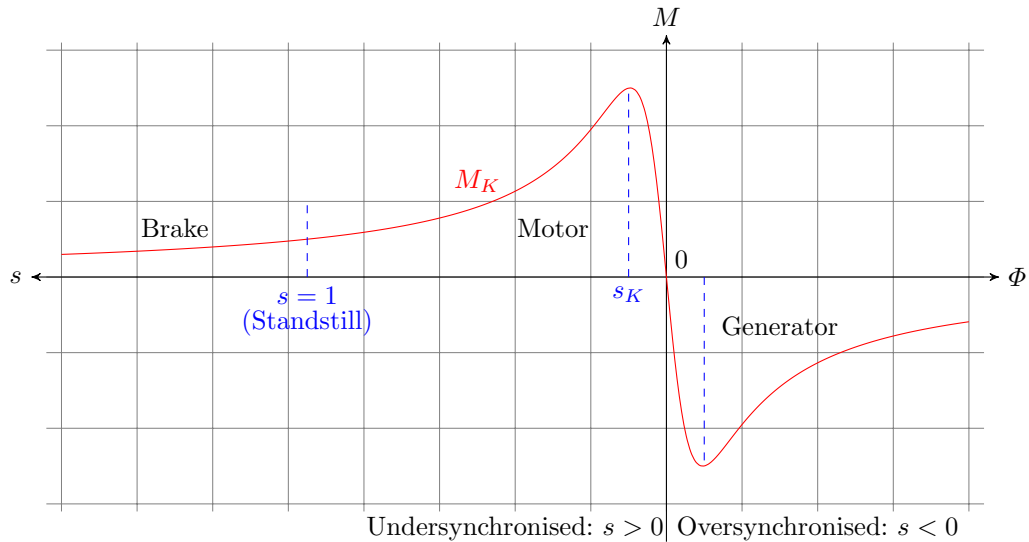


Figure 5.3: **Kloss characteristic curve.** Speed/torque characteristic curve of an asynchronous machine.

Example 5.1: asynchronous machine

A three-phase asynchronous motor has the following information on its nameplate:

- Rated power: 10 kW
- Rated speed: $1440 \frac{\text{U}}{\text{min}}$
- Frequency: 50 Hz
- Breakdown torque: 25%

Calculate the rated torque, the tipping torque and the starting torque.

$$\begin{aligned}
 P_N &= M_N \cdot \omega_N = M_N \cdot 2\pi n_N \\
 M_N &= \frac{P_N}{2\pi n_N} = \frac{10 \cdot 10^3 \text{ W}}{2\pi \cdot 1440 \frac{1}{60 \text{ s}}} = 66,31 \text{ Nm} \\
 s_N &= \frac{\omega_s - \omega_r}{\omega_s} \\
 &= \frac{2\pi \cdot 1500 \frac{1}{60 \text{ s}} - 2\pi \cdot 1440 \frac{1}{60 \text{ s}}}{2\pi \cdot 1500 \frac{1}{60 \text{ s}}} \\
 &= 0,04
 \end{aligned}$$

To calculate the other moments, equation 5.2 is calculated for the cases of nominal operation and standstill.

$$\begin{aligned}M &= \frac{2 \cdot M_K}{\frac{s_K}{s} + \frac{s}{s_K}} \\M_K &= \frac{M_N}{2} \cdot \left(\frac{s_K}{s_N} + \frac{s_N}{s_K} \right) \\&= \frac{66,31 \text{ Nm}}{2} \cdot \left(\frac{0,25}{0,04} + \frac{0,04}{0,25} \right) = 212,54 \text{ Nm} \\M_A &= \frac{2 \cdot 212,54 \text{ Nm}}{\frac{0,25}{1} + \frac{1}{0,25}} = 100,02 \text{ Nm}\end{aligned}$$

A Exercise tasks

A.1 Direct current machine 1

An externally excited DC machine has the following information on its nameplate:

- Rated power: 41,8 kW
- Rated speed: $1900 \frac{1}{\text{min}}$
- Anchor voltage: 440 V
- Anchor current: 100 A
- Excitation : 240 V
- Excitation current: 10 A
- Idling speed: $2000 \frac{1}{\text{min}}$

Note: The rated power on a motor nameplate always refers to the mechanical power output at the rated point.

- a) What is the torque of the machine at the rated point?
- b) Calculate the efficiency of the machine at the rated point.
- c) Calculate the product of the excitation flux and the armature constant. $K \cdot \Phi$
- d) How big is the anchor resistance? R_A ?

B Solutions to the exercises

B.1 Direct current machine 1

- a) The rated torque is directly derived from the rated power and rated speed:

$$P_{\text{mech}} = M \cdot \omega$$

$$M = \frac{P_{\text{mech}}}{\omega} = \frac{41,8 \text{ kW}}{2\pi \cdot 1900 \frac{1}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ s}}} = 210,08 \text{ Nm}$$

- b) In motor operation, the efficiency is the ratio of mechanical power to electrical power. Both the armature power and the excitation power must be taken into account here.

$$\eta = \frac{P_{\text{mech}}}{P_{\text{elektr}}} = \frac{41,8 \text{ kW}}{440 \text{ V} \cdot 100 \text{ A} + 240 \text{ V} \cdot 10 \text{ A}} = 90,08 \%$$

- c) The induced voltage is calculated using equation 3.2. When idling, the induced voltage must correspond to the armature voltage. The same can be seen from equation 3.7, since the torque must also be zero when idling.

$$U_q = K \cdot \Phi \cdot \omega \tag{3.2}$$

$$K \cdot \Phi = \frac{U_q}{\omega} = \frac{440 \text{ V}}{2\pi \cdot 2000 \frac{1}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ s}}} = 2,1 \text{ Vs}$$

- d) There are two possible approaches here. Equation 3.7 can be transformed to the anchor resistance:

$$\begin{aligned}
 n &= \frac{U_A}{2\pi K \cdot \Phi} - \frac{R_A \cdot M_i}{2\pi(K \cdot \Phi)^2} \\
 R_A &= \left(\frac{U_A}{2\pi K \cdot \Phi} - n \right) \cdot \frac{2\pi(K \cdot \Phi)^2}{M_i} \\
 &= \left(\frac{440 \text{ V}}{2\pi \cdot 2,1 \text{ Vs}} - \frac{1900}{60 \text{ s}} \right) \cdot \frac{2\pi \cdot (2,1 \text{ Vs})^2}{210,08 \text{ Nm}} = 220 \text{ m}\Omega
 \end{aligned} \tag{3.7}$$

The other calculation variant uses the equivalent circuit diagram in Figure 3.6. In nominal operation, we can calculate the source voltage U_q . This must be smaller than the source voltage in no-load operation.

$$\begin{aligned}
 U_q &= K \cdot \Phi \cdot \omega \\
 U_{q,n} &= 2,1 \text{ Vs} \cdot 2\pi \cdot 1900 \frac{1}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 418 \text{ V}
 \end{aligned} \tag{3.2}$$

The voltage drop across resistor R_A must be equal to the difference between the armature voltage and the induced voltage through the mesh circulation. Since the armature current is known in nominal operation, the resistance can be calculated using Ohm's law.

$$R_A = \frac{U_A - U_q}{I_A} = \frac{440 \text{ V} - 418 \text{ V}}{100 \text{ A}} = 220 \text{ m}\Omega$$

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